ClearTract™
FOLEY CATHETER

FDA CLEARED

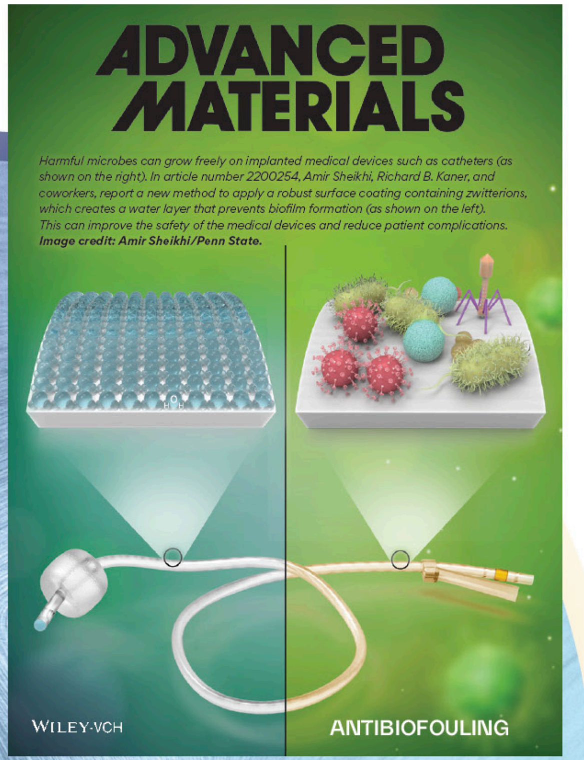
For Suprapubic, Urethral, and Nephrostomy Use

SILQ™
Surface Science Perfected.

THE FIGHT AGAINST CAUTI

CAUTI is a huge and costly problem worldwide.

- There are now an estimated 1 million catheter-associated urinary tract infection (CAUTI) cases per year... in the United States[1] leading to increased morbidity and mortality for patients, costing hospitals up to $10,000/patient to treat.[2]
- "Antimicrobial resistance is an urgent global public health threat, killing at least 1.27 million people worldwide and associated with nearly 5 million deaths in 2019. In the U.S., more than 2.8 million antimicrobial-resistant infections occur each year." - *Antibiotic Resistance Threats in the United States 2019,* CDC[3]
- The Society for Healthcare Epidemiology of America advises against the routine use of antimicrobial/antiseptic impregnated catheters for CAUTI prevention.[4]



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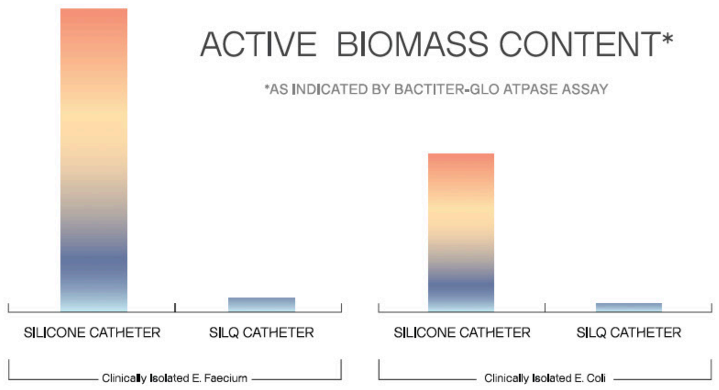
SCAN HERE TO READ

Silq's groundbreaking zwitterion surface technology

For the past twenty years, scientists have shown that bio-inspired polymers known as polyzwitterions strongly resist the deposition of microbes and proteins on surfaces that lead to biofilm formation.[5-15] Silq's proprietary polyzwitterionic surface technology was designed to reduce the risk of infection and other complications associated with implanted medical devices.[16] Read more about this in our *Advanced Materials* article (left).

Silq has been able to successfully apply zwitterionic polymers to medical devices at commercial scale, finally unlocking the tremendous potential of zwitterions. Additionally, Silq's technology does not rely on the use of antibiotic drugs or silver ions that are known to kill microbes in the body. These antimicrobials can lead to patient discomfort[17] and the dangerous proliferation of "superbugs" that are resistant to modern antibiotic treatments.[18-20]

Silq ClearTract™ Catheters

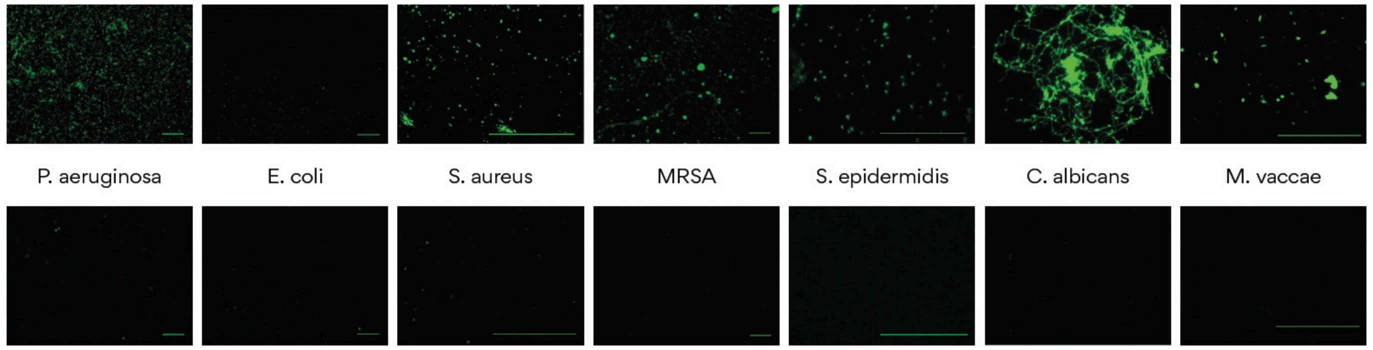


Published data indicate Silq's treatment can suppress microbe adhesion and subsequent biofilm development without the need for antibiotics[16]. Silq's post-market randomized clinical trial (NCT04841226) comparing ClearTract™ to standard and silver-coated catheters is underway across 8 sites in the United States.

From Adhesion to Colonization

When medical devices are implanted, infectious microbes often anchor to the surface and grow into highly resilient biofilms, becoming up to 1000x more resistant to antibiotics.[21,22] Silq's technology addresses this problem by applying a hydrating zwitterionic surface treatment, which can substantially suppress the adhesion of microbes most commonly associated with CAUTI.[16]

UNTREATED SILICONE SURFACE



SILQ™ TREATED SILICONE SURFACE

ClearTract™ Catheters: Improves Patient Comfort and Safety

SILQ'S TREATMENT INCREASES CATHETER LUBRICITY BY UP TO 18X, THEREBY IMPROVING PATIENT COMFORT DURING IMPLANT AND EXPLANT

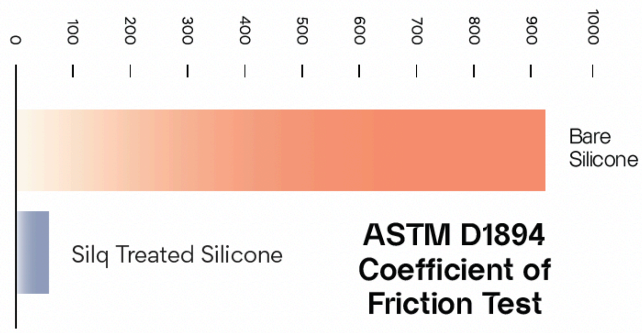
MADE FROM 100% MEDICAL GRADE SILICONE

NO LATEX · **NO BPA** · **NO DEHP**

The days of latex healthcare products are coming to an end - 4.3% of the world population has a sensitivity to latex. Healthcare workers have more than double the latex sensitivity rate due to repeated exposure.[24] ClearTract™ catheters are made from 100% medical grade silicone, protecting both patients and their healthcare providers.

Regarding safety, ClearTract™ catheters have been cleared by the FDA based on the most recent and more rigorous biocompatibility testing requirements for catheters.

DRAG FORCE (GRAMS)



TESTING SUCCESSFULLY COMPLETED

- Cytotoxicity
- Systemic Toxicity
- Irritation
- Pyrogenicity
- Sub-acute Toxicity
- Sensitivity
- Sub-chronic Toxicity
- Genotoxicity*

**Additional requirement specifically for suprapubic clearance*
***All test data available on file*

ClearTract™ Foley Catheter - Ordering Information

CATHETER SIZE	BALLOON SIZE	VALVE	CATHETER SKU#
14F	10cc	2-Way	21400101003
16F	10cc	2-Way	21600101003
18F	10cc	2-Way	21800101003

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1) B Foxman. **The epidemiology of urinary tract infection.** Nat Rev Urol. 2010, 7(12): 653–660. 2) C S Hollenbeak, et al. **The attributable cost of catheter-associated urinary tract infections in the United States: A systematic review.** Am J Infect Control. 2018, 46(7): 751-757. 3) CDC. **Antibiotic Resistance Threats in the United States, 2019.** Atlanta, GA: U.S. Department of Health and Human Services, CDC; 2019. 4) Lo, Evelyn, et al. **"Strategies to Prevent Catheter-Associated Urinary Tract Infections in Acute Care Hospitals: 2014 Update."** Infection Control & Hospital Epidemiology, vol. 35, no. 5, May 2014, pp. 464–479. 5) Singha, Priyadarshini, et al. **"A Review of the Recent Advances in Antimicrobial Coatings for Urinary Catheters."** Acta Biomaterialia, vol. 50, Mar. 2017, pp. 20–40 6) Schleno JB. **Zwitteration: coating surfaces with zwitterionic functionality to reduce nonspeci c adsorption.** Langmuir. 2014 Aug 19;30(32):9625-36. 7) Colilla M, **The Role of Zwitterionic Materials in the Fight against Proteins and Bacteria.** Medicines. 2018 Nov 22;5(4):125. 8) Kazuhiko Ishihara. **Biomimetic materials based on zwitterionic polymers toward human-friendly medical devices,** Science and Technology of Advanced Materials, 23:1, 498-524. 9) Liu S **Recent Advances in Zwitterionic Hydrogels: Preparation, Property, and Biomedical Application.** Gels. 2022; 8(1):46. 10) Yanxian Zhang et al. **Fundamentals and applications of zwitterionic antifouling polymers.** 2019 J. Phys. D: Appl. Phys. 52 403001. 11) Li Q, **Zwitterionic Biomaterials.** Chem Rev. 2022 Dec 14;122(23):17073-17154. 12) Sin, MC. **Hemocompatibility of zwitterionic interfaces and membranes.** Polym J 46, 436–443 (2014). 13) Racovita S. **Polybetaines in Biomedical Applications.** International Journal of Molecular Sciences. 2021; 22(17):9321. 14) Ladd, Jon. **Zwitterionic Polymers Exhibiting High Resistance to Nonspecific Protein Adsorption from Human Serum and Plasma.** Biomacromolecules (2008). 9. 1357-61. 15) Zhang, L., Cao, Z., Bai, T. et al. **Zwitterionic hydrogels implanted in mice resist the foreign-body reaction.** Nat Biotechnol 31, 553–556 (2013). 16) B McVerry, et al. **A Readily Scalable, Clinically Demonstrated, Antibiofouling Zwitterionic Surface Treatment for Implantable Medical Devices.** Adv. Mater. 2022, 34(20): 2200254. 17) Pickard R. **Types of urethral catheter for reducing symptomatic urinary tract infections in hospitalised adults requiring short-term catheterisation: multicentre randomized controlled trial and economic evaluation of antimicrobial- and antiseptic impregnated urethral catheters.** Health Technol Assess 2012;16(47). 18) McNeilly, Oliver, et al. **"Emerging Concern for Silver Nanoparticle Resistance in Acinetobacter Baumannii and Other Bacteria."** Frontiers in Microbiology, vol. 12, 16 Apr. 2021. 19) Hosny, Alaa El-Dien MS, et al. **"The Increasing Threat of Silver-Resistance in Clinical Isolates from Wounds and Burns."** Infection and Drug Resistance, vol. Volume 12, July 2019, pp. 1985–2001. 20) Köves B, Magyar A, Tenke P. **Spectrum and antibiotic resistance of catheter-associated urinary tract infections.** GMS Infect Dis. 2017 06. 21)) W Xu, et al. **Host and bacterial proteases ifluence biofilm formation and virulence in a murine model of enterococcal catheter-associated urinary tract infection.** Biofilms Microbiomes. 2017, 3: 28. 22) D Sharma, et al. **Antibiotics versus biofilm: an emerging battleground in microbial communities.** Antimicrob Resist Infect Control. 2019, 8: 76. 23) M Wu, et al. **Current prevalence rate of latex allergy: Why it remains a problem?**. J Occup Health. 2016, 58(2):138–144.

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